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Modeling water lithium bromide absorption chiller with a heat exchanger in EES and ASPEN Plus

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Abstract

Nowadays energy and its optimized consumption is one of the important problems of human society. In the many of plants most part of input energy wastes. For this reason, utilizing waste energy and improving the thermal efficiency has been suggested for researchers. Absorption chillers can utilize waste energy of industrial plants for providing cooling or refrigeration, so they are able to improve energy efficiency. In present paper, single effect water/lithium bromide absorption chiller with additional heat exchanger in refrigerant side simulated using EES and ASPEN Plus. The modeling is based on steady state condition. The simulating method is explained and the results obtained from two softwares are compared with each other and with published data to verify the simulation.

The results show that COP[1] and exergetic efficiency of the cycle with heat exchanger at refrigerant side are higher than conventional absorption cycle. Also with increasing generator temperature (independent of T_{cond} and T_{evap}), COP and exergetic efficiency first increases and reaches a peak then COP declined slightly and remained relatively constant but exergetic efficiency reduces. In addition, because of the importance of crystallization risk in these systems, the effect of various operating conditions on the possibility of crystallization is investigated too. The results show that probability of crystallization phenomenon increases with increasing the generator temperature.

Keywords: ASPEN Plus, Absorption chillers, Crystallization, Exergy, Heat Exchanger

Introduction

The chillers are one of the most important cooling equipment that they could be generally divided into two categories, compression and absorption chillers. Generally, the compression chillers use electrical energy as main source of energy and the absorption chillers use thermal energy. Due to the high electricity consumption by compression chillers, todays, absorption chillers are well-known among the consulting engineers and owners of residential and office buildings. This type of chillers use thermal energy instead of electrical energy to produce cooling. Their heat source are usually natural gas or gasoline and they have fewer moving parts than compression type and due to the rotational nature of the pumps used in their, failure rates and repair costs are less. They are also much less noise and approximately without vibration than compression type.

The major disadvantage of absorption chillers is low COP that is between 0.5 to 1.5 [1,2] but as the input source of this chillers is usually the waste energy of plants, this disadvantage can be neglected. Also, many researches are performed to improve the performance of absorption cycles in the literature. In this way, in present study a heat exchanger is used in refrigerant side of the cycle.

Absorption chillers in the past have been investigated and modeled by some of ad hoc programs, for example that presented by Lazzarin et al [3]. Studies have also been conducted to develop in-house programs, such as that present by Kim and Infante Ferreira [4] that have been provided analytical models for the absorption cycle with and without circulating refrigerant. As their goal was to evaluate the new working fluid and rapid modeling absorption systems, the obtained equations reduced to second and third order equations. The maximum error for equilibrium temperature of water-lithium bromide solution were evaluated about 1 K. Modern modeling is often done by one of two softwares:

- Absorption Simulation (ABSIM), developed by Oak Ridge National Laboratory [5, 6].
- Engineering Equation Solver (EES), developed at the university of Wisconsin [7, 8].

Users who do modeling in the EES, use the software libraries such as thermo-physical properties of the working fluid. For modeling, all of satisfied equations such as mass conservation, first and second lows of thermodynamics must be given to software. Notice that the number of equations

[1] Coefficient of Performance

must be equals with the number of unknowns. The results of this software have a good accuracy compared with the experimental results [9]. Also work with this software is apprehensible and elimination of defects is easy. However, EES also has limitations, Equations must be defined by the user which makes the modeling complex, tedious and user-prescribed.

ASPEN is a suite softwares of process modeling which is useful. One of this softwares is ASPEN Plus that conduct with the steady-state process modeling [10]. The interface is based on a ready-made libraries, flexible and changeable by user and Fortran-based for components modeling. By linking these components with mass, heat and work flow and input values, the user can model complex processes. ASPEN widely used in the petrochemical industry, this is because of the software capabilities in modeling larger processes using the available properties for the majority of chemical materials that exist in software libraries [11-13]. Graphic-based characteristic of software make modeling easy but despite of the EES, elimination of defects in ASPEN Plus is not easy.

According to topics discussed above, we select EES and ASPEN Plus soft wares for modeling and we compare results with each other.

Description of system

As shown in Figure 1, in this paper, an additional heat exchanger at the refrigerant side is used in absorption refrigeration cycle, the exited refrigerant from the evaporator and the exited saturated liquid from the condenser exchange their heat in this heat exchanger. Saturated liquid gives its heat to the exited refrigerant from the evaporator and enters to the evaporator in a lower temperature. This leads to increase the refrigeration capacity of the chiller and improve the coefficient of performance of the system. In this study since the cooling capacity is assumed constant, the input energy to the system reduces and coefficient of performance increases.

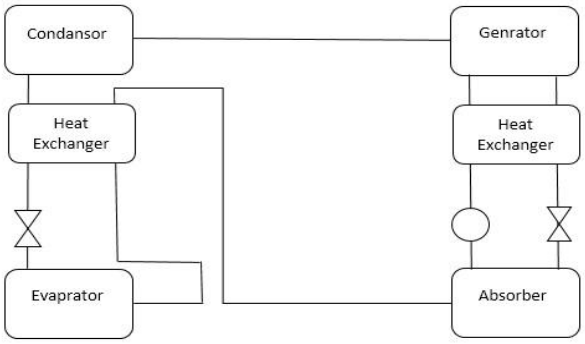


Figure 1. Single effect absorption chiller cycle with heat exchanger at the refrigerant side

Analysis of the system

In order to simulate the systems, some assumptions are made:

- The system is working under steady-state conditions.
- The refrigerant (water) is saturated at the exits of the evaporator and the condenser.
- The solutions are at equilibrium at the exits of the generator and the absorber.
- Pressure losses due to friction in the heat exchangers and in the connecting pipelines are negligible.
- Dissipation of heat does not occur between the system and the surrounding

Energy analysis of absorption systems is possible to apply the two principles that include the law of conservation of mass and first law of thermodynamics. Regardless of the kinetic and potential energy, conservation of mass and the first law of thermodynamics is as follows:

conservation of mass:

$$\sum \dot{m}_i = \sum \dot{m}_e \quad (1)$$

$$\sum \dot{m}_i X_i = \sum \dot{m}_e X_e \quad (2)$$

first law of thermodynamics:

$$\sum \dot{Q} - \sum \dot{W} = \sum \dot{m}_i h_i - \sum \dot{m}_e h_e \quad (3)$$

The coefficient of performance for system defines as folow:

$$COP = \frac{\dot{Q}_{evap}}{\dot{Q}_{gen} + \dot{W}_{pump}} \quad (4)$$

And exergetic efficiency for system defines as folow:

$$\eta_{II} = \frac{\dot{Q}_{evap}(1 - \frac{T_0}{T_{cold}})}{\dot{Q}_{gen}(1 - \frac{T_0}{T_{hot}}) + \dot{W}_P} \times 100 \quad (5)$$

That T_{cold} is the average of inlet and outlet temperature of chilled water at the evaporator, T_{hot} is the average of inlet and outlet temperature of hot water at the generator and T_0 is the surrounding temperature, which define as:

$$T_0 = \frac{(T_{cond} - 8 + T_{cond} - 3)}{2} \quad (6)$$

Verification

For analysis of systems there are several input data that are lists in table 1:

Table 1. inputed data

1	$\dot{Q}_{evap}$(KW)	10.772
2	η_{pump}(-)	0.95
3	T_{evap}(°C)	1.3
4	T_{abs}(°C)	32.7
5	T_{cond}(°C)	40.2
6	T_{gen}(°C)	89.9

In order to ensure the accuracy of the calculations, the results of modeling software in present work are compared with results of Somers and et al [14] for single-effect absorption cycles without heat exchanger that results are listed in table 2:

Table 2. compare present work with Somers[14]

Parameter	Somers (EES)	Present Work		Error (%)	
		EES	Aspen	EES	Aspen
P_{low}(KPa)	0.673	0.6715	0.6715	0.22	0
P_{high}(KPa)	7.445	7.46	7.4606	0.20	0.01
$\dot{Q}_{abs}$ (KW)	14.297	14.09	13.923	1.47	1.18
$\dot{Q}_{cond}$ (KW)	11.427	11.49	11.432	0.55	0.50
$\dot{Q}_{gen}$ (KW)	14.952	14.81	14.592	0.96	1.47
$\dot{Q}_{evap}$ (KW)	10.772	10.772	10.772	0	0
COP	0.720	0.7273	0.738	1	1.47

Resuls and discussion

A heat exchanger at the refrigerant side is utilized to improve the performance of the absorption chiller. We investigate The effect of heat exchanger on COP and exergetic efficiency of the cycle at various operating condition. Also the possibility of crystallization is considered in the cycle.

The effect of added heat exchanger and the generator, condenser and evaporator temperatures on coefficient of performance of the cycle are shown in figures 2 and 3. It can be seen that with increasing generator temperature (independent of T_{cond} and T_{evap}), COP first increases and reaches a peak and then declines slightly and remains relatively constant. So the optimum generator temperature for different conditions can be defined. The maximum coefficient of performance occurs at the lowest temperature of the condenser and the highest temperature of evaporator. Also it is obvious that using refrigerant side heat exchanger improve the COP of the cycle.

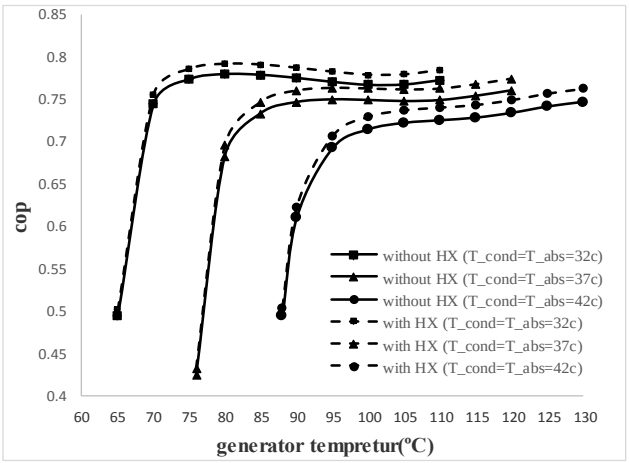


Figure 2. COP vs generator temperature at different condensor temperature

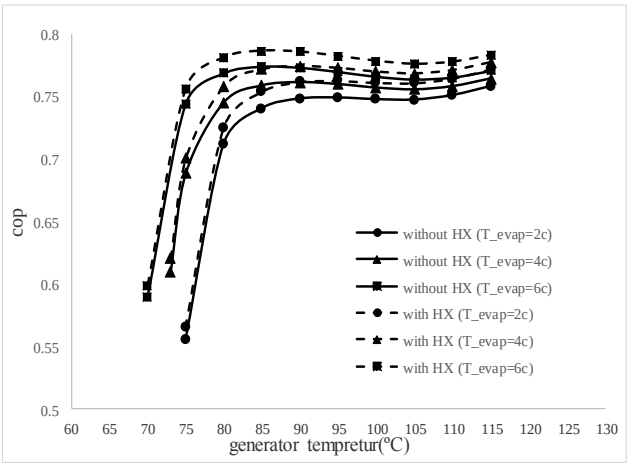


Figure 3. COP vs generator temperature at different evaporator temperature

The effect of added heat exchanger and the generator, condenser and evaporator temperatures on exergetic efficiency of the cycle are shown in figures 4 and 5. It is shown that the exergetic efficiency initially increases with increasing the generator temperature to reach a peak. After this point, increasing the generator temperature reduce the exergetic efficiency with high rate. Figures also show the effect of condenser and evaporator temperature that the maximum exergetic efficiency occurs at the highest temperature of the condenser and the lowest temperature of evaporator.

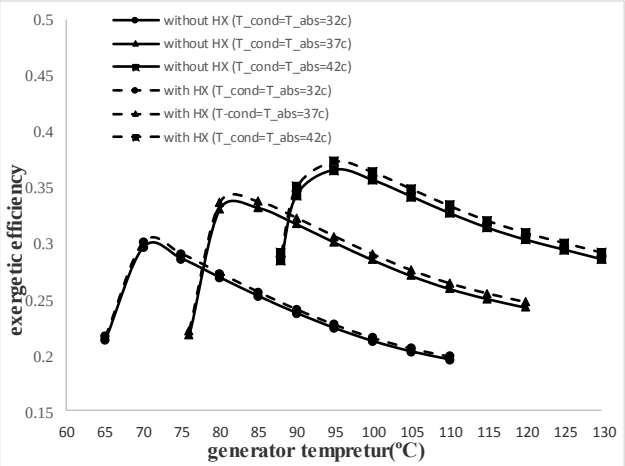


Figure 4 . exergetic efficiency vs generator temperature at various condensor temperatures

Crystallization

The crystallization is one of the important problems in absorption cycles. The actual location of crystallization in cycle can be determined by the mechanical structure of the system. However, studies have shown that the most likely place to crystallization is entry point of solution to the absorber. At this point the solution has the lowest temperature and the highest concentration compared to the other points. To check this phenomenon at entry point of the absorber, concentration and temperature of the point are calculated at various operating conditions. In figure 6, generator temperature range has been determined for the occurrence of this phenomenon. The results are the same for both cycles with and without refrigerant side heat exchanger. It is observed that by increasing the condenser temperature, crystallization occurs at higher generator temperature.

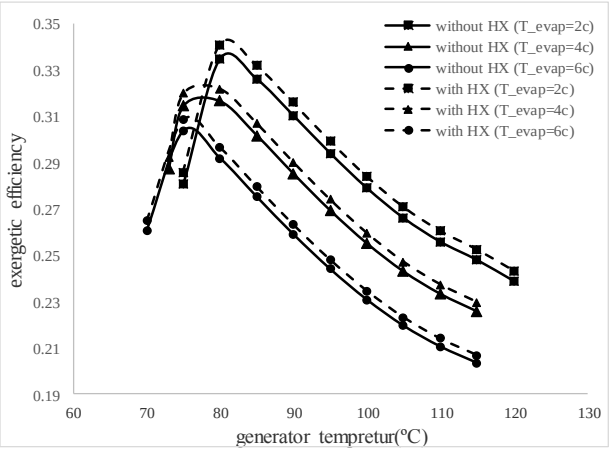


Figure 5. exergetic efficiency vs generator temperature at various evaporator temperatures

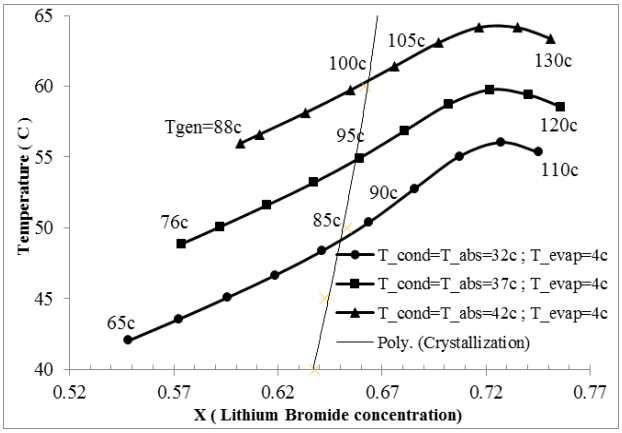


Figure 6. Temperature vs LiBr concentration for investigate of crystallization

Conclusions

The performance of two configuration of single effect absorption chiller (with and without refrigerant side heat exchanger) have been studied and compared.

It is shown that for the two considered systems increasing generator temperature to a certain value increases the COP and exergetic efficiency. Further increases in generator temperature leave the value of COP approximately constant and reduce the value of exergetic efficiency. In all conditions, existence of heat exchanger at the refrigerant side increases COP and exergetic efficiency.

Also investigation of the crystallization phenomenon shows that at each operating condition generator temperature is allowed to increase to a certain value. Further increases in generator temperature lead to crystallization of solution.

Nomenclatures

COP	Coefficient of Performance
$\dot{m}$	Mass
T	Temperature
P	Pressure
$\dot{Q}$	Heat
$\dot{W}$	Work
X	Concentration

Greek symbols

η	efficiency

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